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Calibration results
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Normalized Residuals
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Reprojection error (cam0):     mean 0.2991879818477229, median 0.2706905308098451, std: 0.1766016214032615
Reprojection error (cam1):     mean 0.31277981971936963, median 0.29365241461670377, std: 0.16332637931118954
Gyroscope error (imu0):        mean 0.03373034927953076, median 0.022050594968770886, std: 0.03273529603163909
Accelerometer error (imu0):    mean 0.08609735058565035, median 0.061633566189316 49, std: 0.11806690158888916

Residuals
----------------------------
Reprojection error (cam0) [px]:     mean 0.2991879818477229, median 0.2706905308098451, std: 0.1766016214032615
Reprojection error (cam1) [px]:     mean 0.31277981971936963, median 0.29365241461670377, std:
0.16332637931118954
Gyroscope error (imu0) [rad/s]:     mean 0.28946284538544187, median 0.18923100704372914, std:
0.28092362327250814
Accelerometer error (imu0) [m/s^2]: mean 0.28941343858829266, median 0.20717922447060952, std:
0.3968780425863769

Transformation (cam0):
-----------------------
T_ci:  (imu0 to cam0):
[[ 0.01263097 -0.99987296 -0.00972204  0.00208162]
 [-0.0182368   0.00949085 -0.99978865  0.00019077]
 [ 0.99975391  0.0128056  -0.0181146  -0.00031699]
 [ 0.          0.          0.          1.        ]]

T_ic:  (cam0 to imu0):
[[ 0.01263097 -0.0182368   0.99975391  0.0002941 ]
 [-0.99987296  0.00949085  0.0128056   0.00208361]
 [-0.00972204 -0.99978865 -0.0181146   0.00020522]
 [ 0.          0.          0.          1.        ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
0.000870265006476619


Transformation (cam1):
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T_ci:  (imu0 to cam1):
[[ 0.0267528  -0.99954802 -0.01371309 -0.24768746]
[-0.01195527  0.0133971  -0.99983878 -0.00003437]
[ 0.99957059  0.02691243 -0.01159146  0.00693894]
[ 0.          0.          0.          1.        ]]

T_ic:  (cam1 to imu0):
[[ 0.0267528  -0.01195527  0.99957059 -0.00031004]
[-0.99954802  0.0133971   0.02691243 -0.2477618 ]
[-0.01371309 -0.99983878 -0.01159146 -0.00335049]
[ 0.          0.          0.          1.        ]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
0.006325771651071481

Baselines:
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Baseline (cam0 to cam1):
[[ 0.99989227  0.00373575  0.01419481 -0.24976508]
[-0.00382593  0.99997264  0.00633091 -0.00021516]
[-0.01417077 -0.00638453  0.99987921  0.00728661]
[ 0.          0.          0.          1.        ]]
baseline norm:  0.24987143477713597 [m]


Gravity vector in target coords: [m/s^2]
[ 0.02366331 -9.68243539 -1.55509094]


Calibration configuration
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cam0
-----
```

Camera model: pinhole
Focal length: [1746.4539465334938, 1744.4669421726232]
Principal point: [958.9950621906956, 562.5906820013532]
Distortion model: radtan
Distortion coefficients: [-0.25701113919362983, 0.10262664411100493, 0.00043516222338876745, -2.646838006691349e-05]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.026399999999999996 [m]

cam1

Camera model: pinhole
Focal length: [1744.9386686543533, 1742.546552845488]
Principal point: [939.6720202553483, 569.8432138862356]
Distortion model: radtan
Distortion coefficients: [-0.25645546865320074, 0.1025772034872798, -0.0003740545335494358, -0.00022666394863838954]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.026399999999999996 [m]

IMU configuration
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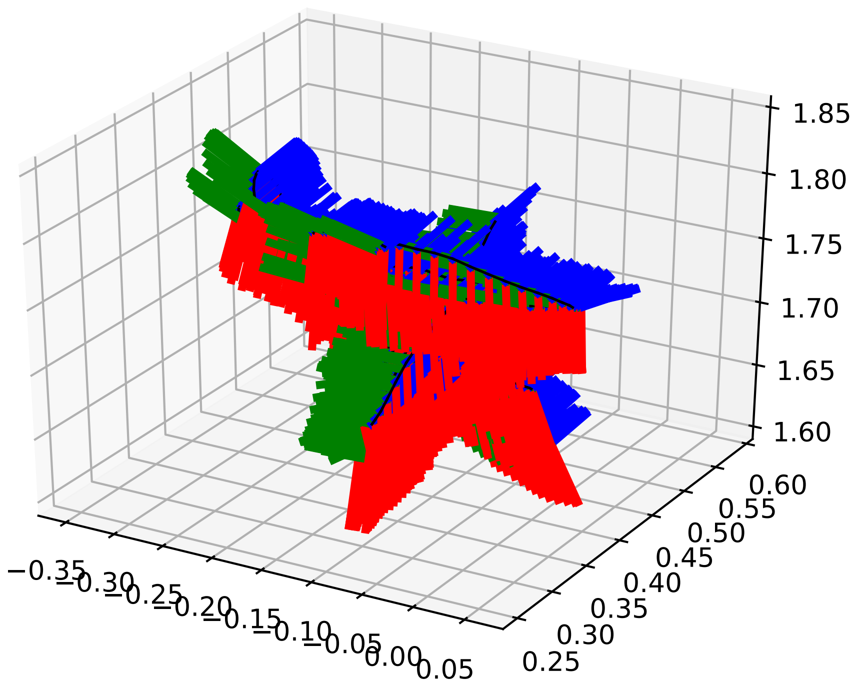
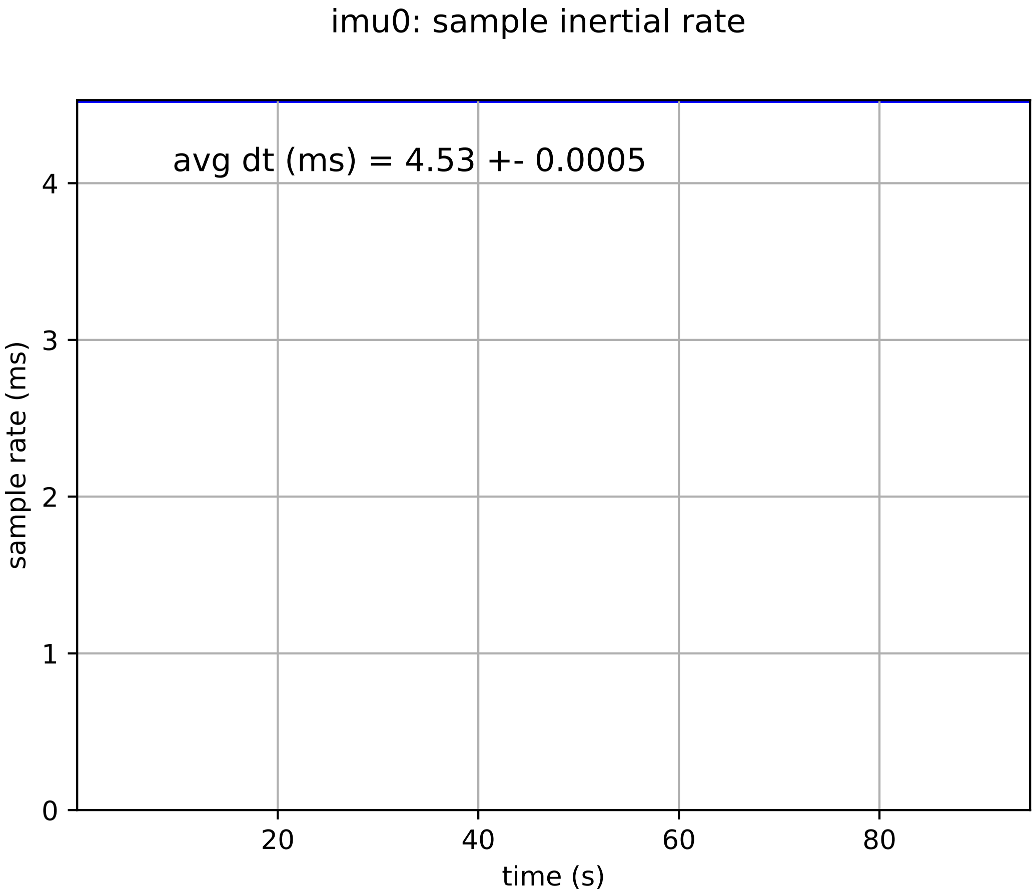
IMU0:

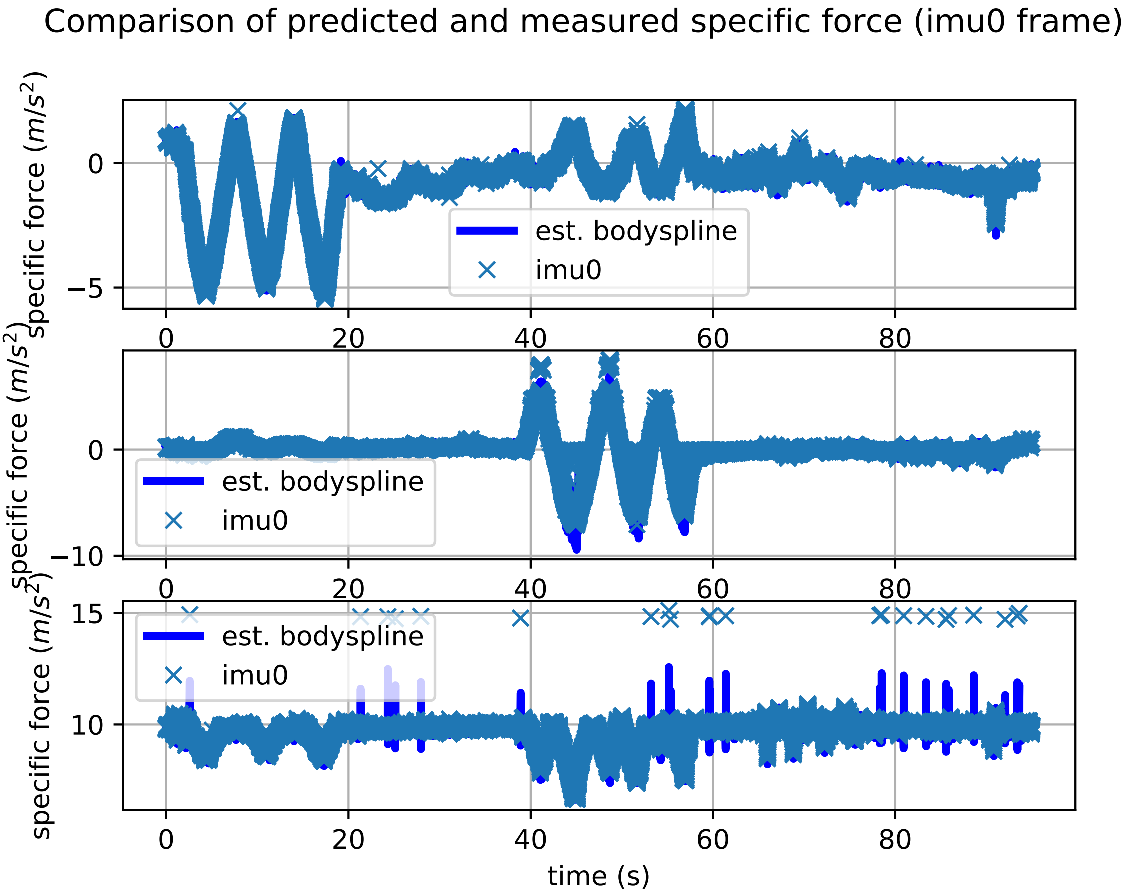
Model: calibrated
Update rate: 208.0

Accelerometer:
Noise density: 0.2330758294
Noise density (discrete): 3.3614674158919886
Random walk: 0.01339530913
Gyroscope:
Noise density: 0.5950319985
Noise density (discrete): 8.581673524534246
Random walk: 0.02296163583
T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

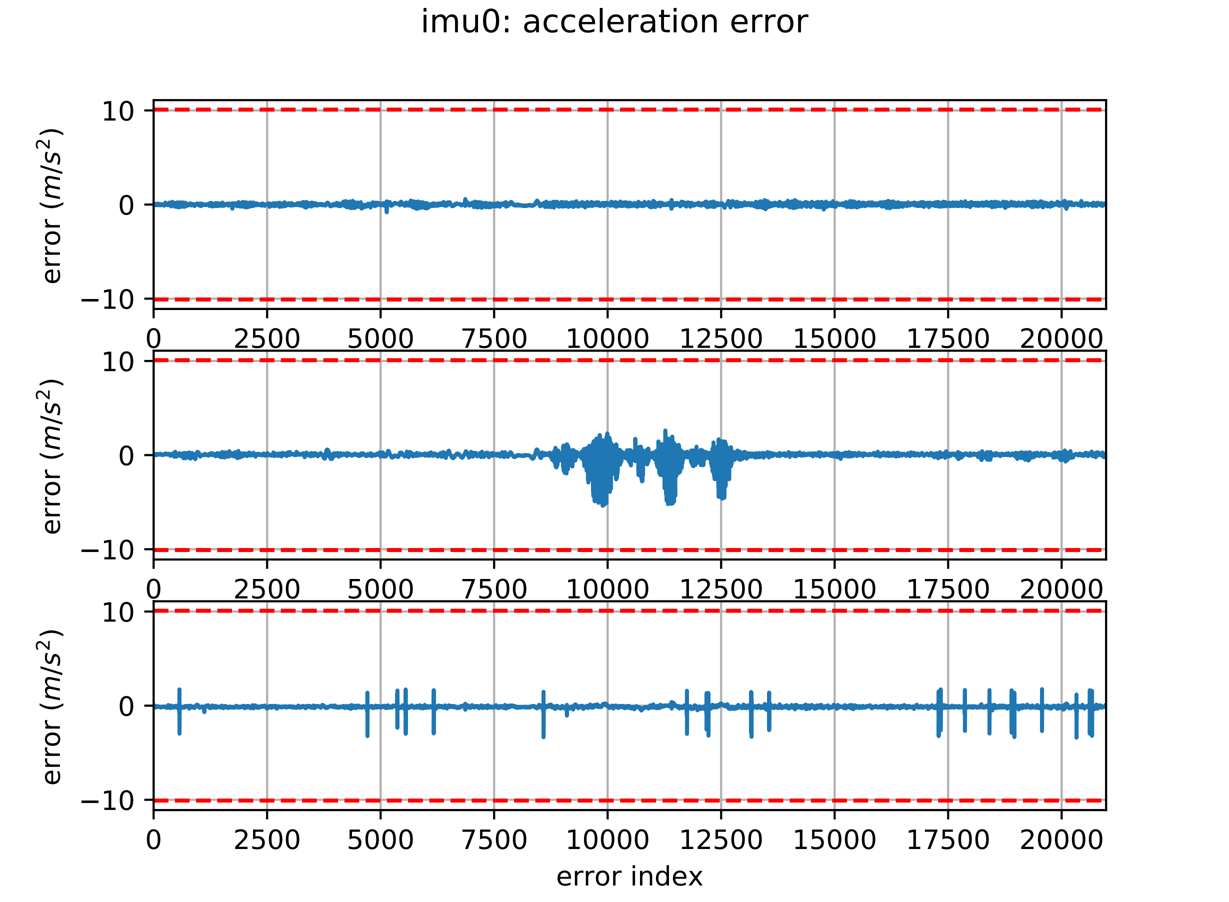
imu0: estimated poses

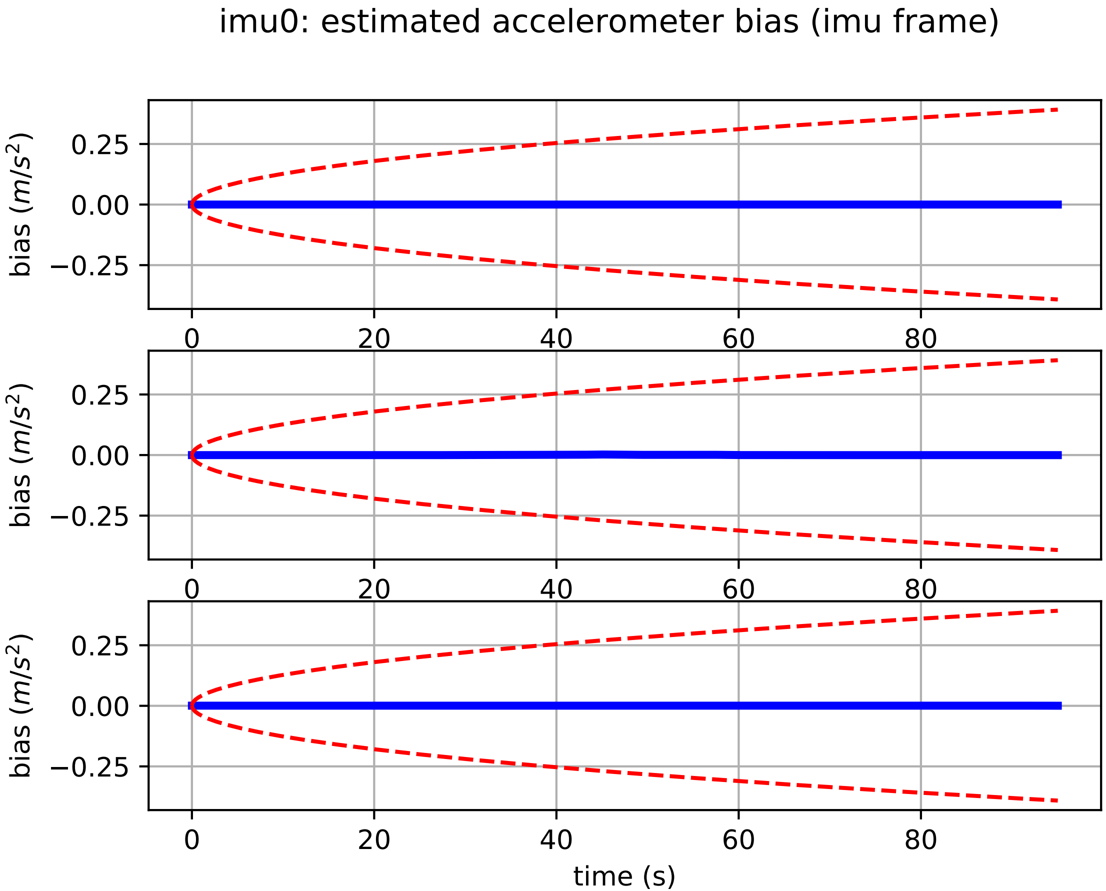


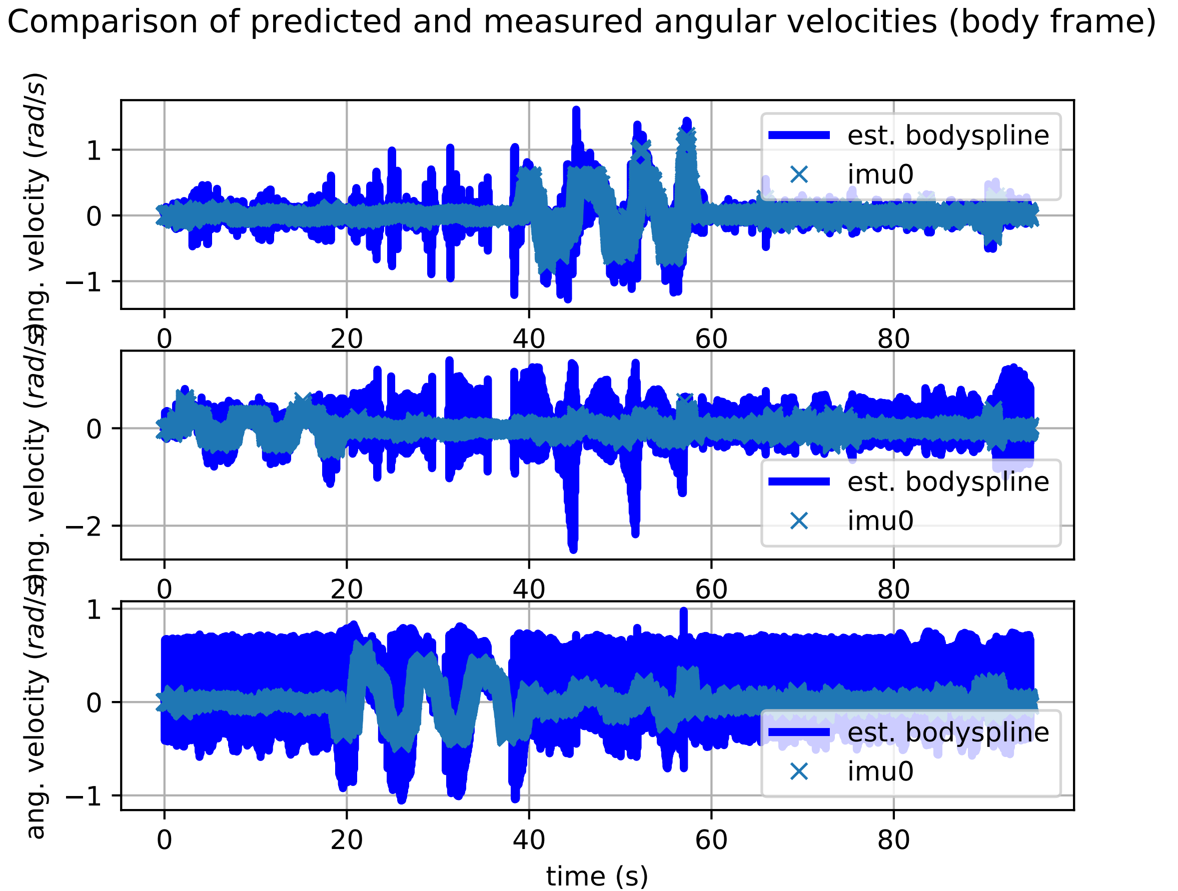

imu0: sample inertial rate
avg dt (ms) = 4.53 +- 0.0005
sample rate (ms)
time (s)
0
1
2
3
4
20
40
60
80

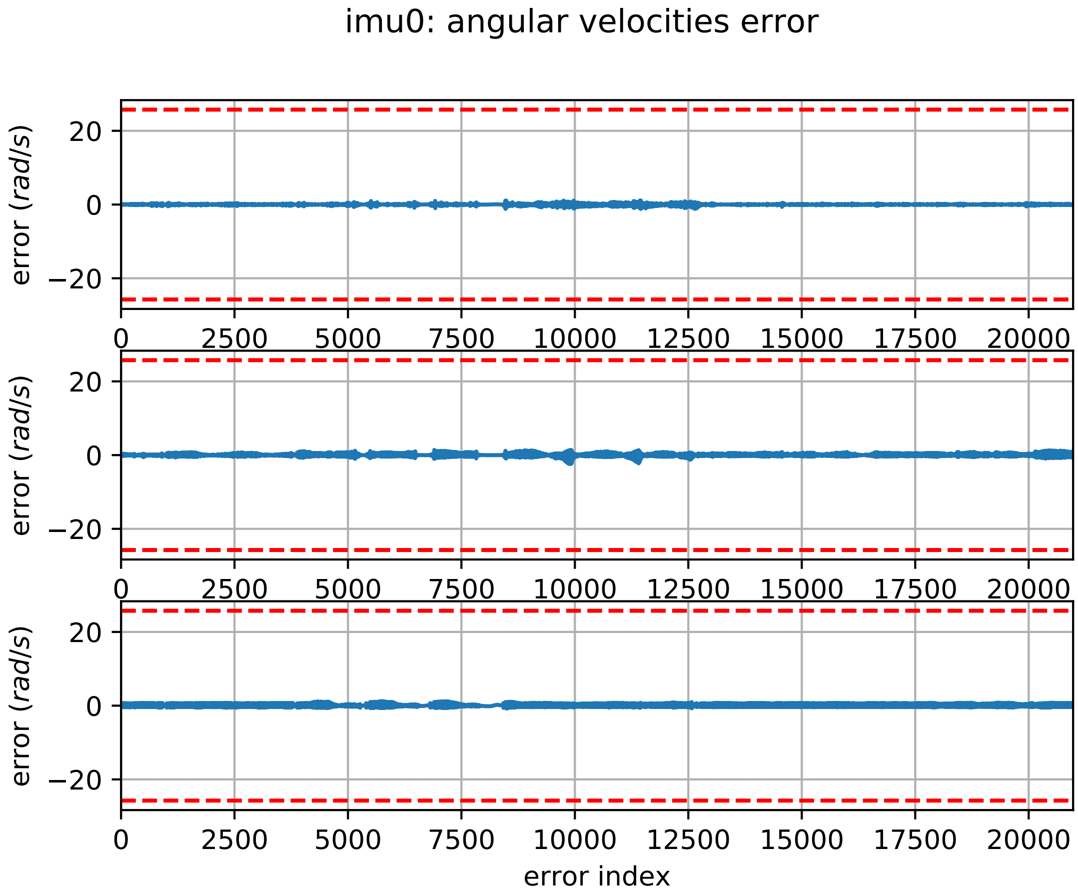


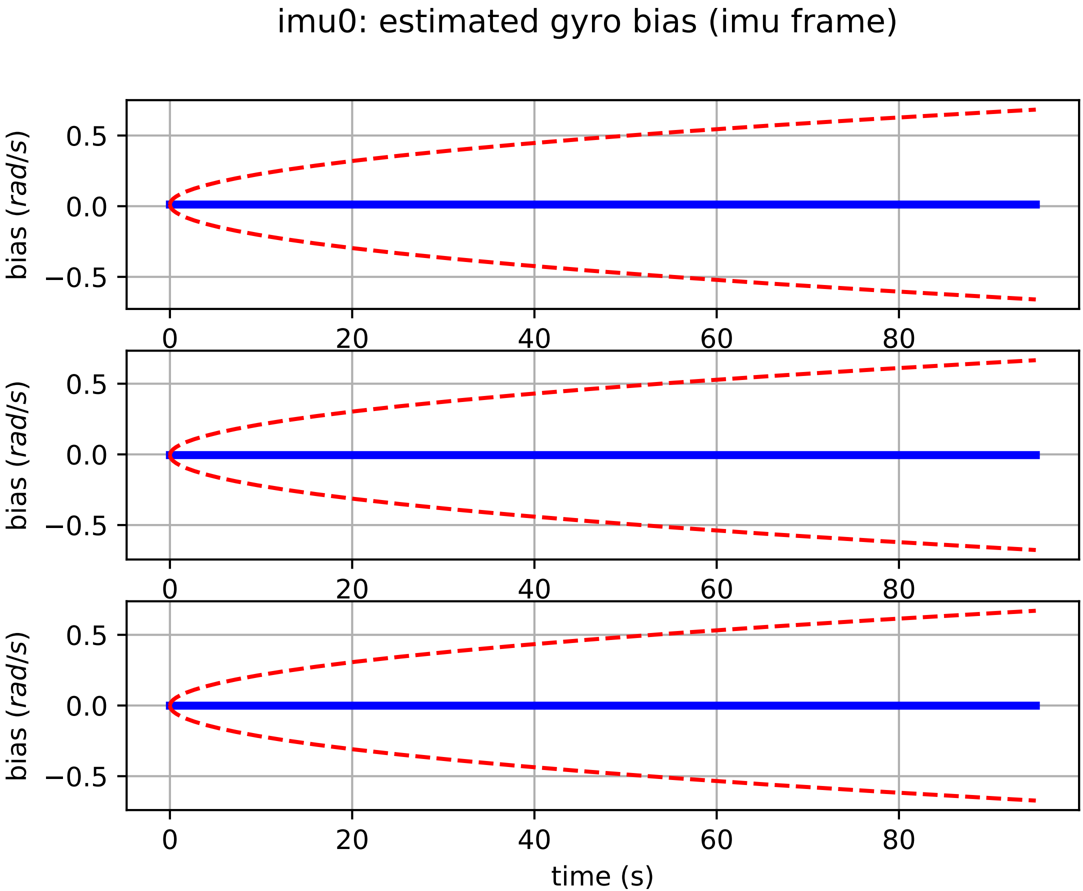
Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
est. bodyspline
imu0
0
−5
0
20
40
60
80
specific force (m/s^2)
est. bodyspline
imu0
0
−10
0
20
40
60
80
specific force (m/s^2)
est. bodyspline
imu0
15
10
0
20
40
60
80
time (s)

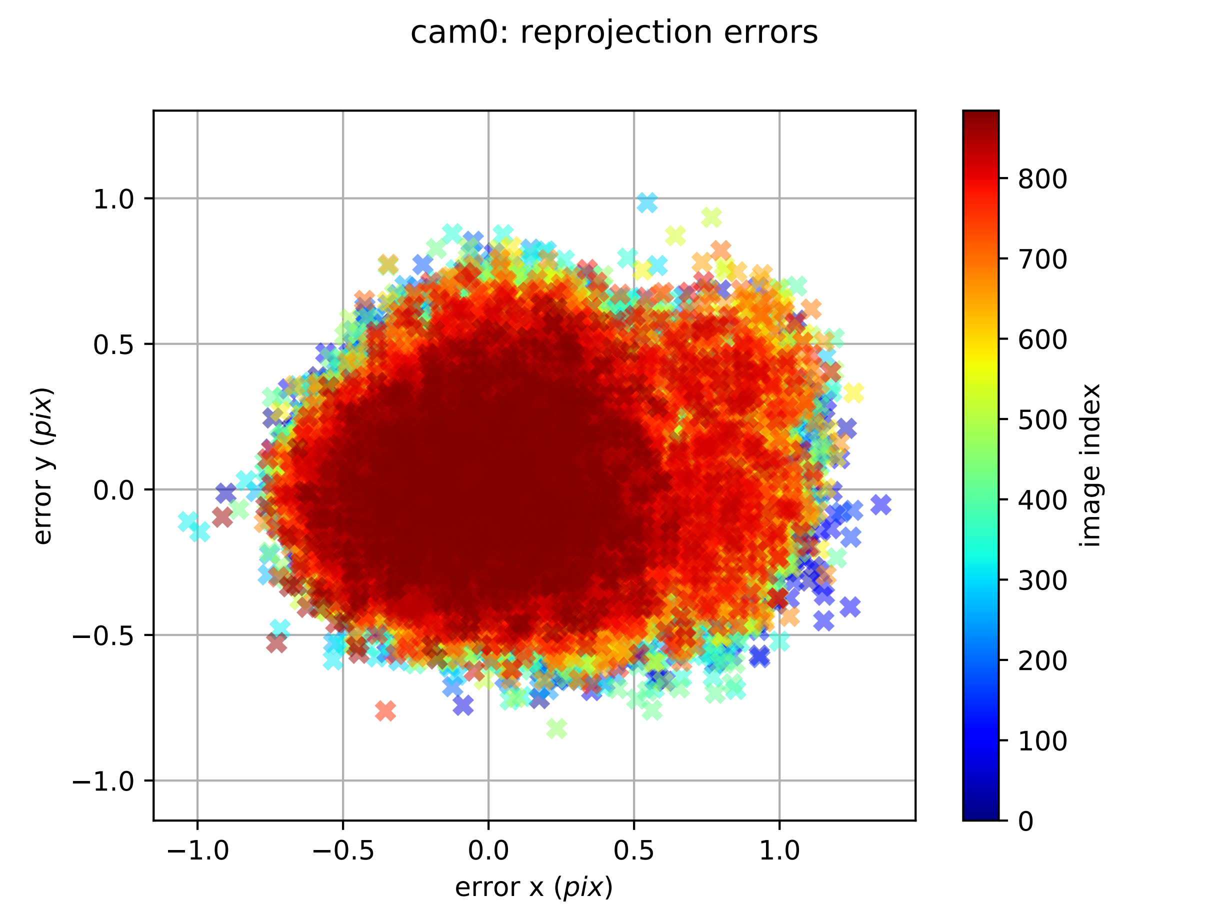

imu0: acceleration error
error (m/s^2)
10
0
−10
0
2500
5000
7500
10000
12500
15000
17500
20000
error index

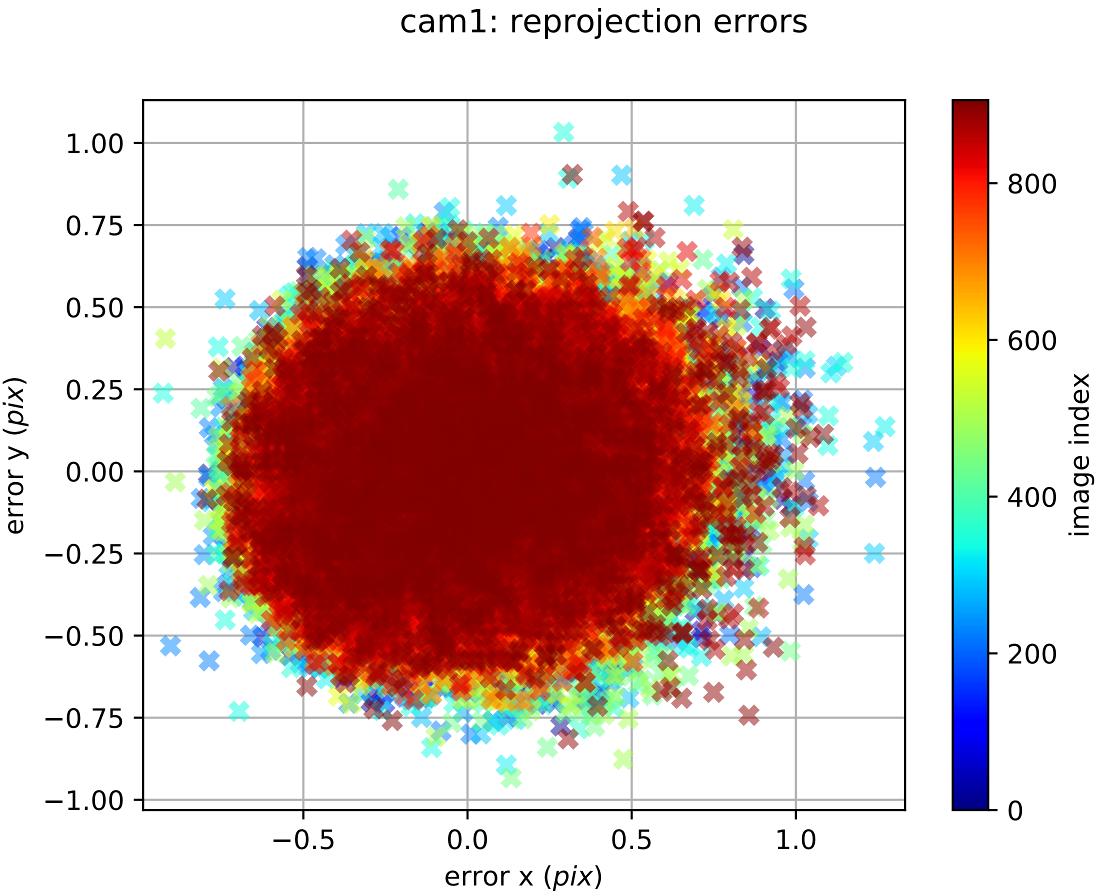

imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.25
0.00
−0.25
0
20
40
60
80
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
est. bodyspline
imu0
1
0
−1
−2
0
20
40
60
80
time (s)


imu0: angular velocities error
error (rad/s)
20
0
−20
0
2500
5000
7500
10000
12500
15000
17500
20000
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
0.5
0.0
−0.5
0
20
40
60
80
time (s)


cam0: reprojection errors
error y (pix)
error x (pix)
image index
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
800
700
600
500
400
300
200
100
0


cam1: reprojection errors
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
error y (pix)
−0.5
0.0
0.5
1.0
error x (pix)
800
600
400
200
0
image index